CRYPTOGRAPHY & CYBER SECURITY
UNIT I, II & III — Complete Theory + Practical Question Bank

Exam-oriented notes | Simple Hinglish explanations with English definitions | Solved numericals

How to use these notes

Har long answer mein definition, working/steps, example, advantages/limitations aur applications likhne ki habit banao. Numerical mein formula, substitution, intermediate steps aur final result clearly show karo. “All possible questions” literally predict nahi kiye ja sakte, lekin neeche syllabus ke main examinable concepts cover kiye gaye hain.

UNIT I — Introduction to Cryptography and Network Security

1. Core concepts and definitions

Q. What is cryptography?

Answer: Cryptography is the science of protecting information by converting readable plaintext into unreadable ciphertext using an algorithm and a key. The authorized receiver uses the appropriate key to recover the plaintext.

Q. What is network security?

Answer: Network security means policies, technologies and controls used to protect networked systems and data from unauthorized access, misuse, modification, disclosure and disruption.

Q. Define plaintext, ciphertext, encryption, decryption, cipher and key.

Answer: Plaintext is the original readable message. Ciphertext is the transformed unreadable message. Encryption converts plaintext to ciphertext. Decryption converts ciphertext back to plaintext. A cipher is the algorithm used for encryption/decryption. A key is a value that controls the algorithm.

Q. What is cryptanalysis?

Answer: Cryptanalysis is the study of recovering plaintext or secret keys without being given the key through weaknesses in the algorithm, implementation or usage.

Q. Differentiate cryptography and cryptanalysis.

Answer: Cryptography designs methods to protect information; cryptanalysis studies methods to break or evaluate those protections.

2. Security goals (CIA triad and supporting goals)

Q. Explain the major security goals.

Answer: Confidentiality ensures that only authorized people can read information. Integrity ensures that data is not altered improperly. Availability ensures that systems and data are accessible when needed. Authentication verifies the identity of a user or the origin of a message. Authorization determines what an authenticated entity is permitted to do. Non-repudiation provides evidence so a party cannot credibly deny a performed action, commonly using digital signatures and audit records.

- Confidentiality example: encrypting a private email.
- Integrity example: detecting an altered bank transaction using a message authentication code.
- Availability example: redundant servers and DDoS protection.
- Authentication example: password plus one-time code.
- Authorization example: a student can view marks but cannot edit them.
- Non-repudiation example: a digitally signed contract with verifiable evidence.

Q. Explain the CIA triad.

Answer: The CIA triad consists of Confidentiality, Integrity and Availability. It is a basic model for security design. A system should prevent unauthorized disclosure, prevent unauthorized or unnoticed modification, and remain accessible to authorized users.

3. Security attacks

Q. What is a security attack? Classify security attacks.

Answer: A security attack is an action that compromises confidentiality, integrity or availability. Passive attacks observe communications without changing them; active attacks alter data, impersonate entities or disrupt services.

- Passive attacks: release of message contents, traffic analysis. They are difficult to detect, so encryption and traffic protection are important.
- Active attacks: masquerade, replay, message modification and denial of service (DoS). They are often addressed using authentication, integrity checks, anti-replay controls, access control and redundancy.

Q. Differentiate passive and active attacks.

Answer: Passive attacks monitor or collect information and do not change system resources; examples are eavesdropping and traffic analysis. Active attacks modify, inject, replay or block data; examples are masquerade, replay, modification and DoS. Passive attacks primarily threaten confidentiality; active attacks can threaten integrity, authenticity and availability.

Q. Explain masquerade, replay, message modification and DoS.

Answer: Masquerade: attacker pretends to be a legitimate entity. Replay: attacker captures a valid message and sends it again later. Message modification: attacker changes, deletes, reorders or delays message content. Denial of Service: attacker makes a service unavailable by exhausting or disrupting resources.

4. Security services and mechanisms

Q. What is a security service?

Answer: A security service is a function provided by a system or protocol to protect data and communication. Examples include authentication, access control, confidentiality, integrity, and non-repudiation.

Q. What is a security mechanism?

Answer: A security mechanism is a method or process used to detect, prevent or recover from an attack. Examples include encryption, digital signatures, access controls, authentication exchange, traffic padding and audit trails.

Q. Differentiate security service and mechanism.

Answer: A service describes WHAT protection is required; a mechanism describes HOW the protection is implemented. Confidentiality is a service; encryption is a mechanism that can provide it.

Q. List common security services.

Answer: Authentication, access control, data confidentiality, data integrity, non-repudiation and availability-related protection.

Q. List common security mechanisms.

Answer: Encipherment/encryption, digital signatures, access control, data integrity checks, authentication exchange, traffic padding, routing control and notarization; auditing and recovery are also practical supporting controls.

5. Security techniques and symmetric-key cryptography

Q. What is symmetric-key cryptography?

Answer: Symmetric-key cryptography uses the same secret key, or closely related shared secret keys, for encryption and decryption. Sender and receiver must securely share the key beforehand.

Q. State advantages and disadvantages of symmetric encryption.

Answer: Advantages: fast, efficient for large amounts of data and relatively low computational cost. Disadvantages: secret-key distribution is difficult, each communicating pair may require key management, and compromise of the shared key exposes protected data.

Q. Differentiate symmetric and asymmetric cryptography.

Answer: Symmetric cryptography uses a shared secret key and is usually fast for bulk data. Asymmetric cryptography uses a public/private key pair and helps with key establishment and signatures but is generally slower. Real systems often use

hybrid cryptography: asymmetric methods establish/authenticate keys and symmetric encryption protects the data.

6. Traditional symmetric ciphers

Q. What is a substitution cipher?

Answer: A substitution cipher replaces each plaintext letter or group with another symbol according to a rule. The positions generally remain in the same order. Examples: Caesar cipher and monoalphabetic substitution.

Q. What is a transposition cipher?

Answer: A transposition cipher rearranges the positions of plaintext characters according to a key or pattern; the characters themselves are not replaced. Examples: rail-fence and columnar transposition.

Q. Differentiate substitution and transposition.

Answer: Substitution changes the identity of symbols while generally preserving their order. Transposition preserves the symbols but changes their positions. Both are traditional techniques and by themselves are not secure for modern use.

7. Caesar cipher — formulas and solved example

Map letters A=0, B=1, …, Z=25. Encryption: C = (P + k) mod 26. Decryption: P = (C – k) mod 26. Here k is the shift.

Example: Encrypt HELLO using k=3. H(7)→K(10), E(4)→H(7), L(11)→O(14), L→O, O(14)→R(17). Ciphertext = KHOOR. Decryption shifts each ciphertext letter back by 3, recovering HELLO.

Example with wrap-around: X with k=3 gives (23+3) mod 26 = 0 = A.

Q. Encrypt CAT using Caesar shift 4.

Answer: C→G, A→E, T→X. Ciphertext: GEX.

Q. Decrypt KHOOR using Caesar shift 3.

Answer: K→H, H→E, O→L, O→L, R→O. Plaintext: HELLO.

8. Monoalphabetic and Playfair ciphers

Q. What is a monoalphabetic substitution cipher?

Answer: Each plaintext letter is consistently mapped to one ciphertext letter using a fixed substitution alphabet. It has a large key space compared with Caesar, but frequency analysis can reveal patterns because letter frequencies are preserved under substitution.

Q. Explain the Playfair cipher.

Answer: Playfair is a digraph substitution cipher: it encrypts pairs of letters using a 5×5 key matrix, usually combining I/J. Prepare plaintext in pairs, insert a filler (often X) between repeated letters in a pair, and pad an odd final letter. Same row: replace each letter with the letter to its right (wrap around). Same column: replace each with the letter below (wrap around). Rectangle: replace each letter with the letter in its row and the other letter's column. Decryption reverses row/column direction.

9. Hill cipher

Q. What is the Hill cipher?

Answer: The Hill cipher encrypts blocks of letters using matrix multiplication modulo 26. Convert letters A=0…Z=25 and use C = K·P mod 26 for column-vector convention. The key matrix must be invertible modulo 26 for decryption.

Small example: K = [[3,3],[2,5]], plaintext HI → P=[7,8]. C=K·P mod 26 = [3×7+3×8, 2×7+5×8] mod 26 = [45,54] mod 26 = [19,2] = TC. For decryption, use the modular inverse matrix K■¹ mod 26.

10. Rail-fence and columnar transposition

Q. Explain rail-fence cipher with example.

Answer: Write plaintext diagonally across a chosen number of rails and read row by row. With 2 rails, plaintext HELLOWORLD can be written alternating between two rows; reading the first row and then second produces the ciphertext. Decryption reconstructs the rail pattern and reads diagonally.

Q. Explain columnar transposition.

Answer: Write plaintext in rows under a keyword, order columns according to the alphabetical order of keyword letters, then read columns in that order. Decryption restores columns in the same keyed order and reads rows.

Unit I — quick revision / likely questions

- Define cryptography, cryptanalysis, plaintext, ciphertext, encryption and decryption.
- Explain CIA triad with examples.
- Classify security attacks and distinguish passive from active attacks.
- Explain security services and mechanisms with examples.
- Compare symmetric and asymmetric cryptography.
- Explain substitution and transposition ciphers and compare them.
- Solve Caesar cipher encryption and decryption numericals.
- Explain monoalphabetic, Playfair, Hill, rail-fence and columnar transposition ciphers.
- Explain replay, masquerade, message modification and DoS attacks.
- Write a note on authentication, access control and non-repudiation.

UNIT II — Modern Symmetric-Key Ciphers

1. Feistel cipher structure

Q. What is a Feistel cipher?

Answer: A Feistel cipher splits a data block into left and right halves. In each round, the new left half is the previous right half and the new right half is the previous left half XORed with a round function applied to the previous right half and round key: L■=R■■■; R■=L■■■ XOR F(R■■■,K■). Decryption uses the same structure with round keys in reverse order.

Q. Why is the Feistel structure important?

Answer: It allows a reversible block cipher to be built even when the round function F itself is not invertible. Security comes from multiple rounds, key scheduling and suitable round functions.

2. S-DES (Simplified DES)

Q. What is S-DES?

Answer: S-DES is a teaching cipher that illustrates the structure of DES using a 10-bit key, an 8-bit plaintext block, two rounds and simplified permutations/substitutions. It is not secure for real-world use.

Typical S-DES key generation: apply P10 to the 10-bit key, split into two 5-bit halves, circular-left-shift each half by one, apply P8 to obtain K1; shift the halves by two more positions and apply P8 to obtain K2. Encryption: initial permutation IP, fk with K1, swap halves, fk with K2, inverse IP. Each fk expands/permutates the right 4 bits, XORs the round key, passes through S-boxes, applies P4, then XORs with the left half.

Q. How is S-DES decrypted?

Answer: Use the same overall structure but apply subkeys in reverse order: K2 first, then K1. For a fully worked exam numerical, carefully apply the specific IP, EP, P10, P8, P4 and S-box tables provided by the course/textbook; these tables must not be guessed.

3. DES

Q. Explain DES algorithm.

Answer: DES is a symmetric block cipher with a 64-bit block and a 56-bit effective key (64 bits including 8 parity bits). It uses an initial permutation, 16 Feistel rounds, a final swap and inverse initial permutation. Each round expands the 32-bit right half to 48 bits, XORs a 48-bit round key, substitutes through eight S-boxes (6 bits to 4 bits each), applies a permutation, and combines with the left half. DES is obsolete for security because its effective key is too short.

Q. Explain DES key generation.

Answer: The 64-bit key is passed through PC-1, which drops parity bits and permutes the remaining 56 bits. Split into two 28-bit halves, rotate left by a round-dependent schedule, then apply PC-2 to select 48 bits for each round key. There are 16 round keys.

Q. Why is DES considered insecure today?

Answer: Its 56-bit effective key is vulnerable to exhaustive key search with modern computing. AES is preferred for modern applications.

4. Double DES and Triple DES

Q. Explain Double DES.

Answer: Double DES applies DES encryption twice with two keys: C=E(K2,E(K1,P)). Decryption is P=D(K1,D(K2,C)). Although the nominal key length is 112 bits, meet-in-the-middle attacks make its effective security much lower than expected, so Double DES is not recommended.

Q. Explain Triple DES (3DES).

Answer: A common form uses Encrypt-Decrypt-Encrypt: C=E(K3,D(K2,E(K1,P))). Decryption reverses the operations: P=D(K1,E(K2,D(K3,C))). With two-key 3DES, K1=K3. 3DES improves on DES but is slow and is being retired in favor of AES.

Q. Compare DES, Double DES and Triple DES.

Answer: DES uses one 56-bit effective key and is insecure. Double DES applies DES twice but is vulnerable to meet-in-the-middle attacks. Triple DES applies DES three times and offers stronger legacy protection, but it is slower and has a 64-bit block size; AES is the modern choice.

5. AES

Q. Explain AES.

Answer: Advanced Encryption Standard is a symmetric substitution-permutation block cipher with a 128-bit block and key sizes of 128, 192 or 256 bits. It uses 10, 12 or 14 rounds respectively. Unlike DES, AES is not a Feistel cipher.

AES encryption stages: initial AddRoundKey; repeated rounds of SubBytes (nonlinear S-box substitution), ShiftRows (row-wise cyclic shifts), MixColumns (column mixing in GF(2■), omitted in the final round), and AddRoundKey (XOR with round key). Key expansion generates round keys. Decryption applies inverse transformations in the corresponding reverse process.

Q. Differentiate DES and AES.

Answer: DES: 64-bit block, 56-bit effective key, 16 Feistel rounds, 48-bit round keys, obsolete. AES: 128-bit block, 128/192/256-bit keys, 10/12/14 rounds, substitution-permutation structure, widely used and considered secure when correctly implemented.

Q. What are the four main AES transformations?

Answer: SubBytes, ShiftRows, MixColumns and AddRoundKey.

6. Block cipher and stream cipher

Q. What is a block cipher?

Answer: A block cipher transforms fixed-size blocks of plaintext into ciphertext under a key. Examples include AES and DES. Modes of operation allow block ciphers to process longer messages.

Q. Differentiate block and stream ciphers.

Answer: A block cipher processes fixed-size blocks; a stream cipher generates or uses a keystream and combines it with data, often bit-by-bit or byte-by-byte. Examples: AES is a block cipher; ChaCha20 is a stream cipher.

7. Modes of operation

Q. What is ECB mode?

Answer: Electronic Codebook encrypts each block independently: C■=E■(P■), and P■=D■(C■). Identical plaintext blocks under the same key create identical ciphertext blocks, revealing patterns. ECB is generally not recommended for messages with repeated structure.

Q. What is CBC mode?

Answer: Cipher Block Chaining XORs each plaintext block with the previous ciphertext block before encryption. C■=E■(P■ XOR IV); C■=E■(P■ XOR C■■■). Decryption: P■=D■(C■) XOR C■■■, using IV for the first block. IV should be unpredictable and unique as required by the scheme.

Q. What is CFB mode?

Answer: Cipher Feedback turns a block cipher into a self-synchronizing stream-like mode. It encrypts the previous ciphertext (or IV), then XORs the output with plaintext to produce ciphertext. It can operate on smaller segments.

Q. What is OFB mode?

Answer: Output Feedback repeatedly encrypts an internal feedback value to create a keystream; plaintext is XORed with the keystream. Errors in a ciphertext bit generally affect only the corresponding plaintext bit, but reusing the same key/IV pair is dangerous.

Q. What is CTR mode?

Answer: Counter mode encrypts successive counter blocks to generate a keystream, then XORs it with plaintext: C■=P■ XOR E■(counter■). It supports parallel processing and random access. A counter/nonce must never repeat with the same key.

Q. Compare ECB, CBC, CFB, OFB and CTR.

Answer: ECB encrypts blocks independently and leaks repeated patterns. CBC chains plaintext blocks with previous ciphertext and requires an IV. CFB feeds ciphertext into the block cipher and is self-synchronizing. OFB generates a keystream from feedback and bit errors do not propagate far. CTR encrypts counters, supports parallelism and random access, and requires unique counter/nonce values.

8. Mode numericals (symbolic worked examples)

For actual numerical work, the question must provide a cipher/key or simplified toy cipher. Without E■ values, a real AES/DES ciphertext cannot be calculated by hand. The following symbolic examples show the exact XOR steps used in exams.

CBC example with 4-bit toy values: P1=1010, IV=1100. First XOR: P1 XOR IV = 0110. If the given toy cipher says E■(0110)=1001, then C1=1001. For P2=0011, compute P2 XOR C1 = 1010; if E■(1010)=0111, then C2=0111. Use the encryption lookup/table supplied in the question.

CTR example: if the toy cipher gives E■(counter1)=1011 and P1=1100, then C1=P1 XOR keystream=0111. Decryption repeats the same XOR: C1 XOR 1011 = 1100. Never reuse a counter/nonce with the same key in real systems.

9. Key Distribution Center (KDC)

Q. What is a KDC?

Answer: A Key Distribution Center is a trusted third party that helps users establish or obtain shared symmetric session keys. Users may each share a long-term secret with the KDC; the KDC distributes a temporary session key protected under those secrets.

Q. Explain the basic KDC process.

Answer: A requests a session key to communicate with B. The KDC creates or selects session key Ks, sends Ks to A protected using A’s long-term key, and provides a ticket for B protected using B’s long-term key. A forwards B’s ticket; B recovers Ks and the parties can communicate securely. Real protocols add nonces, timestamps and authentication to resist replay.

Q. Advantages and limitations of KDC.

Answer: It reduces the need for every pair of users to maintain a separate long-term key and centralizes key management. However, it is a high-value trusted target, can become a bottleneck or single point of failure, and must be strongly protected.

Unit II — likely questions

- Explain Feistel network with round equations and diagram.
- Explain S-DES key generation and encryption/decryption steps.
- Explain DES structure, rounds, S-boxes and key schedule.
- Why is DES insecure? Explain Double DES and meet-in-the-middle attack.
- Explain Triple DES EDE process.
- Explain AES structure, transformations and key sizes.

- Compare DES and AES; block cipher and stream cipher.
- Explain ECB, CBC, CFB, OFB and CTR modes with formulas and comparison.
- Solve a CBC/CTR XOR numerical using the given toy-cipher outputs.
- Explain KDC and session-key distribution with steps.

UNIT III — Mathematics for Cryptography

1. Modular arithmetic

Q. Define modular arithmetic.

Answer: Modular arithmetic works with remainders after division by a positive integer n. a ≡ b (mod n) means n divides a–b, or a and b have the same remainder when divided by n.

Examples: 17 mod 5 = 2; 29 ≡ 5 (mod 12); (7+9) mod 10 = 6; (7×8) mod 10 = 6.

Q. State the basic modular rules.

Answer: Addition: (a+b) mod n = [(a mod n)+(b mod n)] mod n. Subtraction: (a–b) mod n can be reduced modulo n. Multiplication: (ab) mod n = [(a mod n)(b mod n)] mod n. Exponentiation can be reduced repeatedly. Division is not automatically allowed; use a modular inverse only when it exists.

Q. What is a modular multiplicative inverse?

Answer: The inverse of a modulo n is a number x satisfying ax ≡ 1 (mod n). It exists if and only if gcd(a,n)=1.

2. Euclidean algorithm

Q. State the Euclidean algorithm.

Answer: For integers a>b>0, repeatedly divide a by b: a=bq+r, where 0≤r<b. Replace (a,b) with (b,r) until remainder becomes zero. The last non-zero remainder is gcd(a,b).

Solved example: Find gcd(56,24). 56=24×2+8; 24=8×3+0. Therefore gcd(56,24)=8.

Solved example: Find gcd(91,35). 91=35×2+21; 35=21×1+14; 21=14×1+7; 14=7×2+0. Therefore gcd(91,35)=7.

Solved example: Find gcd(252,105). 252=105×2+42; 105=42×2+21; 42=21×2+0. Therefore gcd=21.

Q. Why is the Euclidean algorithm useful in cryptography?

Answer: It efficiently computes greatest common divisors, helps determine whether modular inverses exist, and is used in RSA key generation and modular arithmetic.

3. Extended Euclidean algorithm

Q. What is the Extended Euclidean algorithm?

Answer: It calculates gcd(a,n) and expresses it as ax+ny=gcd(a,n). If gcd(a,n)=1, the coefficient x gives the modular inverse of a modulo n after reducing x modulo n.

Solved inverse example: Find inverse of 7 modulo 26. Euclidean steps: 26=3×7+5; 7=1×5+2; 5=2×2+1. Back-substitute: 1=5–2×2=5–2(7–5)=3×5–2×7=3(26–3×7)–2×7=3×26–11×7. Thus –11×7 ≡1 (mod 26), so inverse is –11 mod 26 = 15. Check: 7×15=105 ≡1 (mod 26).

Solved inverse example: Find inverse of 3 modulo 11. 11=3×3+2; 3=1×2+1. Thus 1=3–(11–3×3)=4×3–11. Inverse is 4 because 3×4=12 ≡1 (mod 11).

Q. Find the inverse of 10 modulo 17.

Answer: 17=1×10+7; 10=1×7+3; 7=2×3+1. Back-substitute: 1=7–2(10–7)=3×7–2×10=3(17–10)–2×10=3×17–5×10. Hence inverse is –5 mod 17 = 12. Check: 10×12=120 ≡1 mod 17.

4. Fermat’s Little Theorem

Q. State Fermat’s Little Theorem.

Answer: If p is prime and p does not divide a, then a^(p–1) ≡ 1 (mod p). Equivalently, a^p ≡ a (mod p) for any integer a.

Solved example: Find 3^100 mod 7. Since 7 is prime and gcd(3,7)=1, 3^6≡1 mod 7. 100=6×16+4, so 3^100≡3^4=81≡4 mod 7. Answer: 4.

Solved example: Find 2^20 mod 13. 2^12≡1 mod 13; 20=12+8. 2^20≡2^8=256; 256 mod 13 = 9. Answer: 9.

Q. When can Fermat's theorem be used?

Answer: Use it when the modulus is prime and the base is not divisible by that prime; verify these conditions before reducing exponents.

5. Euler's theorem and Euler phi function

Q. Define Euler's phi function.

Answer: $\phi(n)$ counts the positive integers from 1 through n that are relatively prime to n. For a prime p, $\phi(p)=p-1$. For distinct primes p and q, $\phi(pq)=(p-1)(q-1)$.

Q. State Euler's theorem.

Answer: If gcd(a,n)=1, then $a^{\phi(n)} \equiv 1 \pmod n$. It generalizes Fermat's Little Theorem to composite moduli.

Solved example: Find 3^20 mod 10. Since gcd(3,10)=1 and $\phi(10)=4$, 3^4≡1 mod 10. 20=4×5, so 3^20≡1 mod 10. Answer: 1.

Solved example: Find $\phi(12)$. Integers 1…12 relatively prime to 12 are 1,5,7,11. Therefore $\phi(12)=4$.

Q. Differentiate Fermat's and Euler's theorem.

Answer: Fermat uses a prime modulus p and exponent p−1 for a not divisible by p. Euler uses any positive modulus n when gcd(a,n)=1 and exponent $\phi(n)$. Fermat is a special case of Euler's theorem for prime n.

6. Chinese Remainder Theorem (CRT)

Q. State the Chinese Remainder Theorem.

Answer: If moduli n1,n2,…,nk are pairwise coprime, then the simultaneous congruences x≡a1 (mod n1), …, x≡ak (mod nk) have a unique solution modulo N=n1n2…nk.

Solved example: x≡2 (mod 3), x≡3 (mod 5). Numbers congruent to 2 mod 3 are 2,5,8,11,14,…; among these, 8 gives remainder 3 when divided by 5. Therefore x≡8 (mod 15).

CRT formula method: N=product of moduli. Ni=N/ni. Find Mi as the inverse of Ni modulo ni. Then $x=\Sigma(ai \cdot Ni \cdot Mi) \bmod N$.

Solved formula example: x≡2 (mod 3), x≡3 (mod 5), x≡2 (mod 7). N=105. N1=35, N2=21, N3=15. Inverses: 35 mod 3=2, inverse 2; 21 mod 5=1, inverse 1; 15 mod 7=1, inverse 1. x=(2×35×2)+(3×21×1)+(2×15×1)=140+63+30=233. 233 mod 105=23. Check: 23 mod 3=2, mod 5=3, mod 7=2. Answer x≡23 mod 105.

Q. Why is CRT useful in cryptography?

Answer: CRT can speed up RSA private-key operations by computing modular exponentiation separately modulo the prime factors and recombining results. It also solves systems of congruences.

7. Unit III practice questions with answers

Q. Find 43 mod 6.

Answer: 43=6×7+1, so answer 1.

Q. Compute (17+28) mod 9.

Answer: 45 mod 9=0.

Q. Compute (13×17) mod 10.

Answer: 221 mod 10=1.

Q. Find gcd(84,30).

Answer: 84=2×30+24; 30=1×24+6; 24=4×6. gcd=6.

Q. Find gcd(119,34).

Answer: 119=3×34+17; 34=2×17. gcd=17.

Q. Find inverse of 5 modulo 12.

Answer: 5×5=25≡1 mod 12, so inverse is 5.

Q. Find inverse of 8 modulo 15.

Answer: 8×2=16≡1 mod 15, so inverse is 2.

Q. Find 2^10 mod 7.

Answer: 2^3≡1 mod 7? 8≡1, so 2^10=2^9×2≡2. Answer 2.

Q. Find $\phi(15)$.

Answer: $\phi(15)=(3-1)(5-1)=8$.

Q. Solve x≡1 mod 4, x≡2 mod 3.

Answer: Numbers 1 mod 4: 1,5,9,…; 5 mod 3=2. So x≡5 mod 12.

Exam writing templates

Template for a 5–8 mark theory answer

- Write a clear definition in 2–3 lines.
- Explain the purpose and main idea.
- Write steps/working or draw a labelled diagram where relevant.
- Give a small example.
- Write advantages, limitations or applications.
- End with a one-line conclusion.

Formula sheet — revise before the exam

- Caesar encryption: C=(P+k) mod 26; decryption: P=(C–k) mod 26.
- Euclidean algorithm: a=bq+r; last non-zero remainder = gcd(a,b).
- Modular inverse: ax≡1 (mod n), exists iff gcd(a,n)=1.
- Fermat: a^(p–1)≡1 (mod p), p prime and p■a.
- Euler: a^ϕ(n)≡1 (mod n), if gcd(a,n)=1.
- CRT: x=Σ(ai·Ni·Mi) mod N, with N=∏ni, Ni=N/ni, Mi=Ni■¹ mod ni.
- Feistel: L■=R■■■; R■=L■■■ XOR F(R■■■,K■).
- CBC: C■=E■(P■ XOR IV); C■=E■(P■ XOR C■■■).
- CTR: C■=P■ XOR E■(counter■).
- DES: 64-bit block, 56-bit effective key, 16 rounds.
- AES: 128-bit block; key sizes 128/192/256 bits; rounds 10/12/14.

Last-day revision checklist

- Can I define all key terms without looking?
- Can I distinguish security goals, attacks, services and mechanisms?
- Can I encrypt/decrypt Caesar cipher and explain substitution vs transposition?
- Can I explain Feistel, S-DES, DES, 3DES and AES structure?
- Can I compare all five modes of operation and write their formulas?
- Can I calculate GCD using Euclidean algorithm and modular inverse using Extended Euclid?
- Can I apply Fermat and Euler theorem only when conditions are satisfied?
- Can I solve a two- or three-congruence CRT question and verify the result?

- For each long answer, can I include definition, steps, example and conclusion?

Important accuracy note

Modern DES/3DES/AES hand calculations require many fixed tables and byte operations. In university exams, use the tables and exact toy values provided by the question or prescribed textbook. Do not invent S-DES permutation/S-box tables or claim a real AES ciphertext without carrying out the full specified algorithm. For practical security, use modern authenticated encryption (for example AES-GCM) through trusted libraries rather than implementing ciphers manually.